Design and Analysis of Aerodynamic and Thermal Solutions for Sidepods in a Student-Designed Formula SAE Vehicle

Team Members: Ryan Lee, Benson Lin, Brandon Oliveira, Aidan Kraus
Faculty Advisor: Professor Emilie Dressaire

1 Abstract

This project centers on the design and optimization of the 2024-2025 Gaucho Racing Formula SAE car's sidepods. This component is critical for two objectives: managing external airflow to reduce drag and control tire wake, and directing airflow towards the car's radiators to cool the vehicle's internal electronics. Using 2D and 3D Computational Fluid Dynamics simulations (CFD), we will analyze the effects of different design variations on drag, downforce, and cooling efficiency. The primary goal is to identify an optimal sidepod geometry that balances these aerodynamic and thermal demands to establish a strong foundation and understanding for future Gaucho Racing designs.

2 Overview

2.1 Background

Formula SAE (FSAE) is an intercollegiate engineering design competition that tasks students with the comprehensive research, design, and manufacture of a formula-style electric race car. The program culminates in an annual competition where teams are evaluated in a series of static and dynamic events that judge both the vehicle's engineering design and its on-track performance. These include static events that judge engineering design, cost analysis, and a business presentation, as well as dynamic events like acceleration, skidpad, autocross, and the critical endurance race. Success in this competition demands a holistic engineering approach, balancing raw performance with efficiency, reliability, and manufacturability. Moreover, the design of the sidepods is a critical subsystem for competitive success. The sidepods serve a dual purpose, being essential for ensuring the thermal reliability of the powertrain while also managing and supporting the vehicle's overall aerodynamic performance.

2.2 Objectives

This project will optimize the sidepod design by managing competing requirements of aerodynamic efficiency and thermal management throughout the car. The sidepods' primary function is to house the radiators that cool the electric drivetrain, a role critical for maintaining optimal operating temperatures in the motor, controller, and battery, especially during endurance events. Pushing these components beyond their thermal limits results in immediate power throttling or a complete shutdown. Therefore, the design must prioritize maximizing the mass flow rate of cooling air directed to the radiator cores, especially at the lower average speeds seen on a tight autocross or endurance style track. The radiator cores use the provided air flow to remove heat from the internal coolant which in turn cools the car. This objective is not just about the inlet size; it involves carefully designing the internal ducting to minimize pressure drop, prevent flow separation, and ensure the velocity profile at the radiator face is as uniform as possible. Wasted areas of the radiator core have the penalty of being dead weight. However, this thermal priority must be balanced with aerodynamic objectives. The external geometry will be shaped to minimize overall drag and to manage the turbulent wake from the front tires, diverting it away from sensitive aerodynamic surfaces. The core challenge is to achieve a design that satisfies all cooling targets with a minimal penalty to drag and weight. The final design must also integrate with the existing chassis and be feasibly manufacturable.

3 Methodology

3.1 Preliminary Analysis and Simulations

Our methodology is built on an iterative process, alternating between 3D modeling in SolidWorks and multi-physics simulation in ANSYS. We will run two distinct analyses in parallel: one for external aerodynamics and another for internal thermal-fluid dynamics. The external aerodynamic analysis will use 3D CFD (ANSYS Fluent) on various sidepod geometries. The primary metrics here will be the total vehicle drag coefficient (C_d) and how well the outwash manages the front tire wake. In parallel, a separate set of 3D CFD simulations will focus on the internal ducting. By modeling the radiator core as a flat plane, we can accurately determine the mass flow rate of air and the total pressure drop across the system, our key indicators of cooling performance. This design process is a feedback loop. An inlet shape that is optimized in the internal thermal analysis will then be integrated into the external aerodynamic model to quantify its drag penalty. Conversely, an aggressive external outwash shape will be tested in the internal simulation to see if it compromises the inlet's efficiency. This iterative trade off study is essential for converging on a design that is optimal across both domains.

3.2 Mounting Solutions

The sidepod mounting system must be structurally sound, capable of supporting the radiators while withstanding significant aerodynamic loads and track-induced vibrations. All designs will be attached to mounts welded onto the chassis, prioritizing minimal weight and serviceability. We will employ Finite Element Analysis (FEA) in ANSYS to verify the structural integrity of all mounting brackets, ensuring the design will not fail under racing conditions. This analysis will also include a study on the feasibility of using Dzus quarter-turn fasteners versus conventional trick tabs.

3.3 Full Car Simulation

As a final validation, the optimized sidepod design will be integrated into a comprehensive, full-vehicle 3D simulation. This model will include all other major aerodynamic elements (front wing, rear wing, undertray, and side pods) to analyze the systemic interactions. Critically, this final simulation will also include a Conjugate Heat Transfer (CHT) analysis. This high-fidelity simulation models the complete thermal pathway: from the powertrain components to the cooling fluid, and from the fluid to the air via the radiators. This will provide a full system prediction of component temperatures under a simulated endurance event load, definitively validating that our design is both aerodynamically efficient and thermally reliable for competition.

4 Project Plan/Addendum

4.1 Ryan Lee

Ryan Lee will serve as the lead designer, responsible for all 3D Computer-Aided Design (CAD) modeling and structural validation within SolidWorks. His primary task is to translate the team's simulation data and functional requirements into a cohesive, manufacturable component. This iterative role involves sculpting the external surfaces for aerodynamic analysis and engineering the complex internal ducting pathways for the thermal-fluid analysis. As simulation data becomes available from the analysis leads, Ryan will refine the parametric model, which serves as the "single source of truth" for the entire project. This central design must not only meet performance targets but also integrate seamlessly with existing chassis hardpoints and other aerodynamic elements.

A second, critical component of his role is the design of all mounting solutions. The sidepods are structural components that must support the full weight and fluid load of the radiators while withstanding significant aerodynamic forces (both pressure and suction) and high-frequency vibrations from the track. He will design all mounting brackets and hardpoints, focusing on minimal weight, structural integrity, and serviceability. This design phase will include a formal trade study on the feasibility of using Dzus quarter-turn fasteners

versus conventional trick tabs, balancing rapid serviceability against weight and structural simplicity.

Finally, Ryan will be responsible for all Finite Element Analysis (FEA) on these components. He will apply realistic load cases to the sidepod structures and mounting brackets. His deliverable will be a formal analysis report verifying that all designs meet a target factor of safety, in addition to a complete Bill of Materials (BOM) and technical data package (TDP) for the manufacturing lead. His work provides the essential CAD framework that integrates the team's simulation and manufacturing efforts.

4.2 Benson Lin

Benson Lin will lead all high-fidelity 3D Computational Fluid Dynamics (CFD) analysis for the project using ANSYS Fluent. His responsibilities are divided into two parallel simulation streams that form the core of the project's validation. First, he will conduct the external aerodynamic analysis on the various SolidWorks geometries provided by Ryan. This involves creating a high-quality, polyhedral mesh with appropriate inflation layers to accurately capture the boundary layer. He will utilize the SST k-omega turbulence model to quantify the total vehicle drag coefficient (Cd) and generate detailed pressure maps. More importantly, this simulation will be used to visualize the effectiveness of the outwash design in managing the front tire's turbulent wake and its subsequent interaction with the rear wing, a key factor in overall vehicle aerodynamic performance.

Second, Benson will perform the internal thermal-fluid analysis. This separate 3D simulation will model the internal ducting and treat the radiator core as a flat plane, using experimentally derived coefficients to represent its flow resistance. This analysis will determine the mass flow rate of air through the core and the total pressure drop across the system—the primary indicators of cooling efficiency. Benson's analysis is central to the team's iterative feedback loop, as he will provide this quantitative data to the design lead. As a final validation, Benson will be responsible for integrating the finalized sidepod design into a comprehensive, full-vehicle 3D simulation. This will include a Conjugate Heat Transfer (CHT) analysis, which couples the fluid dynamics with solid-body heat transfer. This high-fidelity simulation provides a full-system prediction of component temperatures under load, definitively validating the design's performance.

4.3 Brandon Oliveira

Brandon Oliveira is responsible for conducting the 2D CFD analysis that enables the team's rapid design iteration. His work in ANSYS focuses on simulating 2D cross-sections of the sidepod ducts and radiator inlets, which provides a fast, computationally inexpensive method for optimizing fundamental geometries. Treating the problem as two-dimensional allows for the rapid testing of dozens of geometric arrangements at a fraction of the computational cost of a full 3D model. His primary focus will be on the geometry of the inlet duct, optimizing the expansion angle to maximize pressure recovery while minimizing flow separation. He will be analyzing key performance indicators such as the coefficient of pressure (Cp) at the duct inlet, the degree of flow separation, and the uniformity of the velocity profile at the radiator face.

The data from his 2D analysis will serve as the baseline for the 3D modeling and simulation leads. His contribution is essential for narrowing the design scope and ensuring that only the most promising concepts are moved forward to the more time-consuming 3D analysis. He will work closely with the design lead, Ryan, to integrate his findings, allowing Ryan to further develop the 3D geometry from 2D profiles that are already validated. He will also support the 3D CFD lead, Benson Lin, by providing a validated starting point and initial conditions, which allows the 3D analysis to be a tool for validation rather than exploration, saving significant project time. His findings will be compiled into a formal trade study report, creating a valuable repository of data on duct performance sensitivity that will benefit future Gaucho Racing design cycles.

4.4 Aidan Kraus

Aidan Kraus will serve as the manufacturing lead, responsible for translating the team's final digital designs into a high-quality, track-ready physical part. His contribution is entirely focused on the hands-on fabrica-

tion, assembly, and integration of the sidepods. This process begins with taking the final CAD models from the design lead and developing a comprehensive production plan. He will be responsible for preparing and creating the sidepod molds from high-density foam or tooling board. This will be done through using a hot wire cutter to cut the foam layers into shape following wooden profiles, and stacking them up in a method similar to a riemann sum. Following this, he will meticulously prepare these molds, a multi-step process involving sealing, sanding, and adding putty to ensure a high-quality surface finish.

Additionally, he will be leading the composite layups. He will manage the composite materials, ensuring proper storage and handling, and will lead the layup, vacuum bagging, and curing cycle. He will carefully follow material datasheets to achieve the correct resin-to-fiber ratio and avoid common manufacturing defects like voids or delamination. After curing and demolding, he will be responsible for all post-processing, which includes trimming the cured parts to their final shape, performing precision drilling for mounting hardware, and executing the final surface finishing. This role provides the essential, practical grounding for the entire project. Throughout the design phase, Aidan will provide critical "Design for Manufacturability" (DFM) feedback to the design and simulation leads. This feedback ensures that the proposed CAD designs are practical for the team's tools and fabrication capabilities, preventing costly redesigns or late-stage manufacturing failures. Finally, he will lead the physical integration of the sidepods onto the chassis.

5 Timeline

